

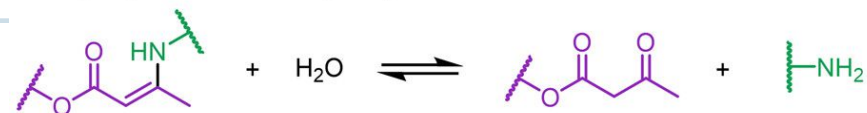
# Covalent Adaptable Networks

## Vinylogous Urethane Transamination (I)

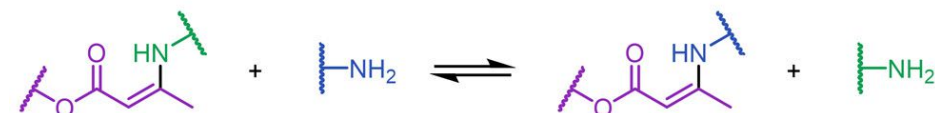
Vinylogous urethanes are not carbamate (urethane) functional groups.

But they behave like electronically extended urethane analogues via **vinylogy** (electronic effects “relayed” through conjugation)

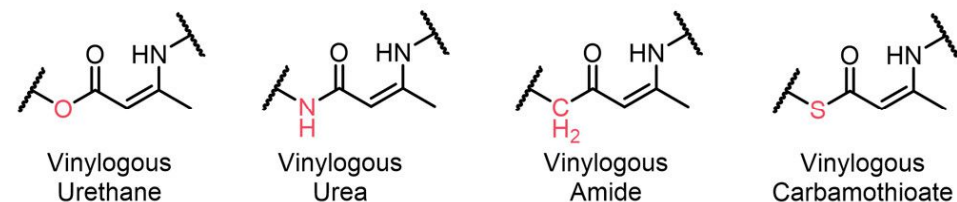
### A. Vinylogous Urethane Hydrolysis



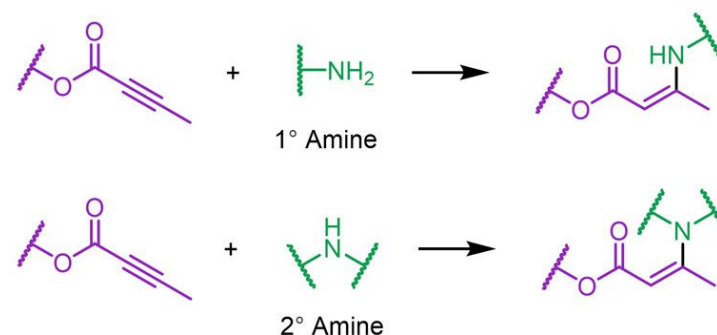
### B. Transamination of Vinylogous Urethane



### C. Vinylogous Acyl Groups with Different Adjacent Atoms



### D. Vinylogous Urethane Synthesis with Alkyne Esters



*Chem Rev.* **2022**, *13*, 370

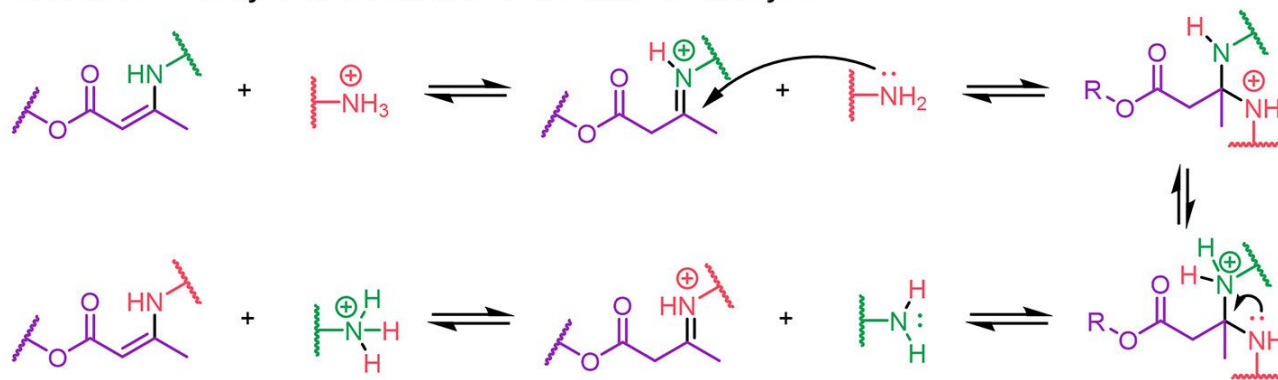
# Covalent Adaptable Networks

## Vinylogous Urethane Transamination (II)

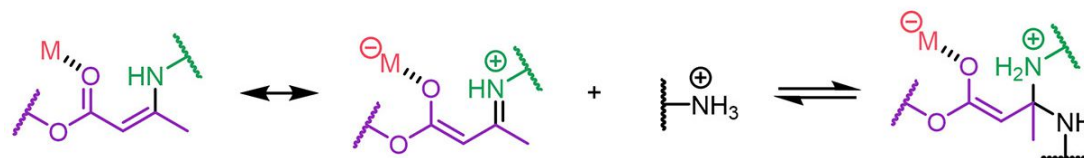
Associative exchange with

- acids (Brønsted or Lewis)
- Brønsted bases

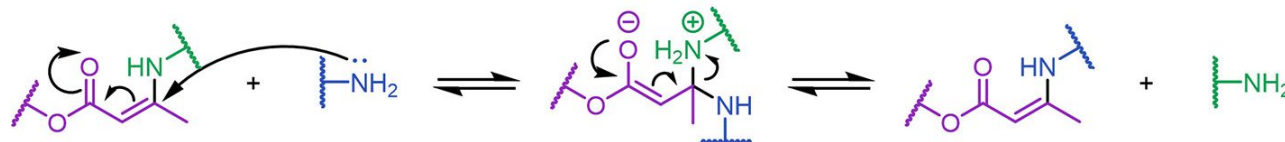
A. Iminium Pathway of Transamination with Protic Acid Catalysts



B. Transamination Catalyzed by Lewis Acids



C. Michael Addition Pathway in the Presence of Base



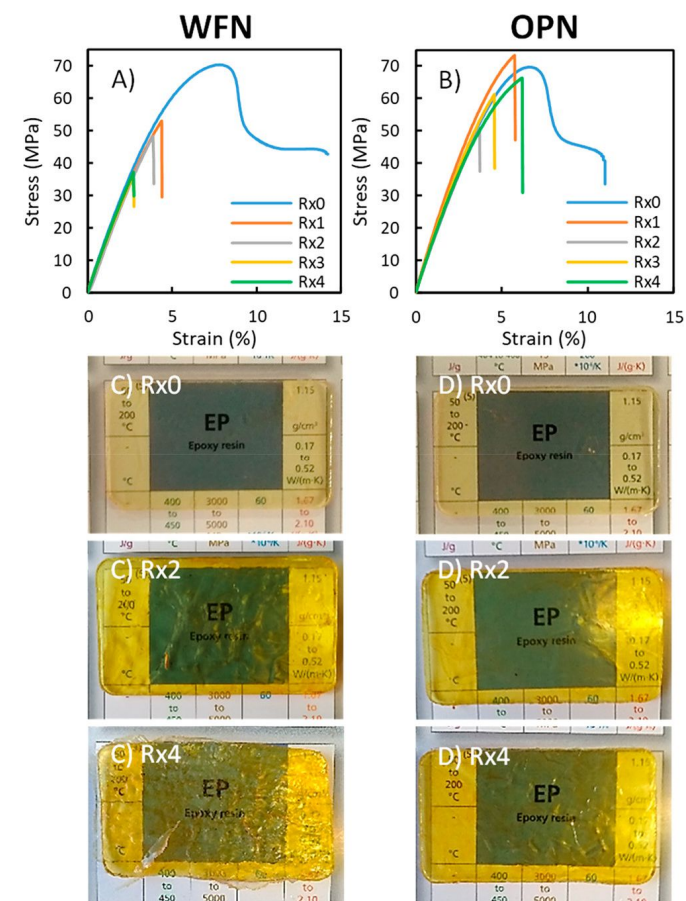
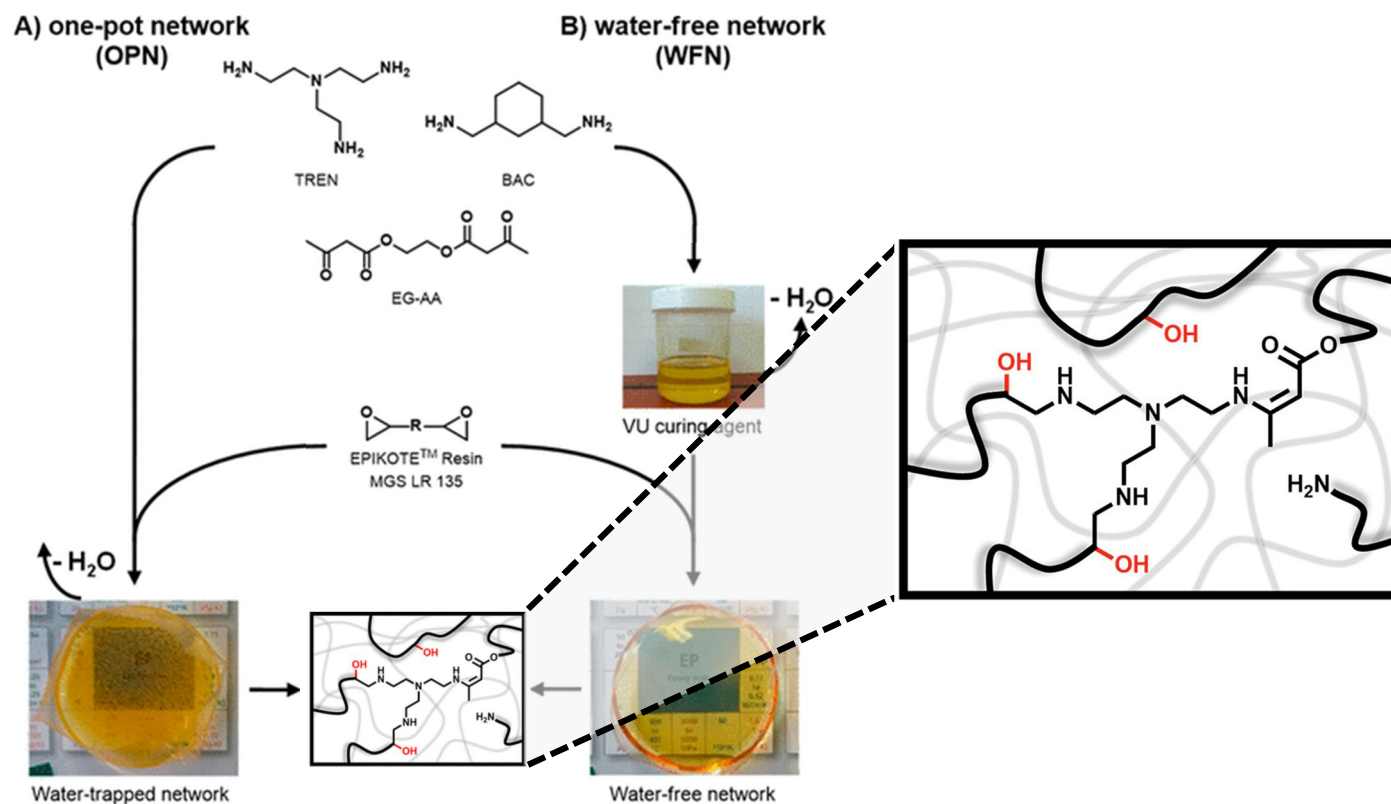
*Chem Rev.* **2022**, *13*, 370

# Covalent Adaptable Networks

## Vinylogous Urethane Transamination (III)

Example: *Epoxy resin*

with exchangeable vinylogous urethane crosslinks



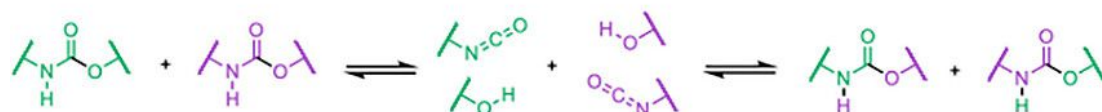
Macromolecules 2020, 53, 2485

# Covalent Adaptable Networks

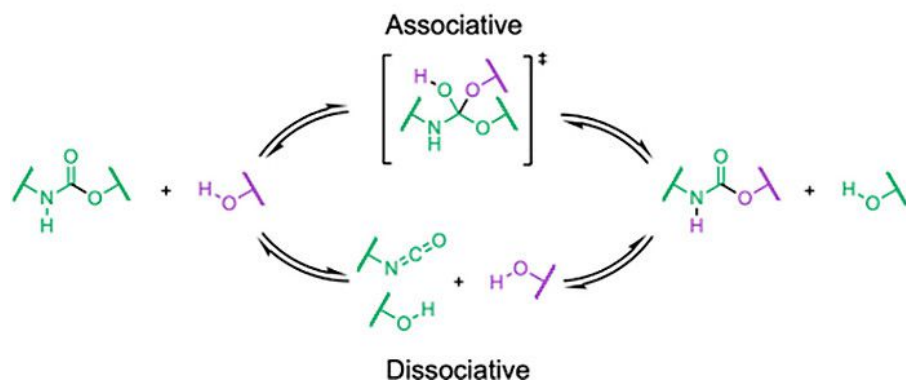
## Urethane CANs (I)

### Urethane Exchange vs Transcarbamylation

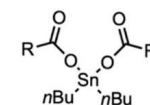
#### A. Urethane Exchange



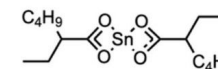
#### B. Transcarbamylation



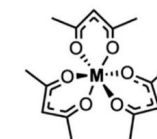
#### Metal-Based Lewis Acids



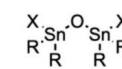
Dibutyltin dilaurate, R = C<sub>11</sub>H<sub>23</sub>  
Dibutyltin maleate, R =



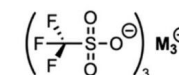
Tin(II) 2-ethylhexanoate



**Metal** acetylacetonate  
where M = Al, Mn, Fe, Cu, Zn, Hf, In

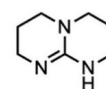


R = nBu, X = OTf, AcO, Br, NCS

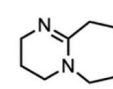


**Metal** trifluoromethanesulfonate  
where M = Sc, Fe, La, Bi, Sm

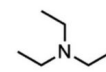
#### Organic Bases



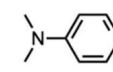
TBD



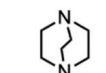
DBU



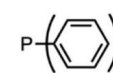
NEt<sub>3</sub>



DMAP

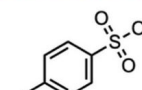


DABCO

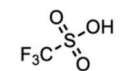


PPh<sub>3</sub>

#### Brønsted Acids



*p*-Toluenesulfonic acid



trifluoromethanesulfonic acid

#### Inorganic Bases

KOH

LiOH

*t*-BuOK

NaOH

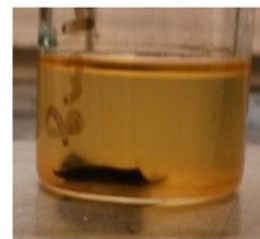
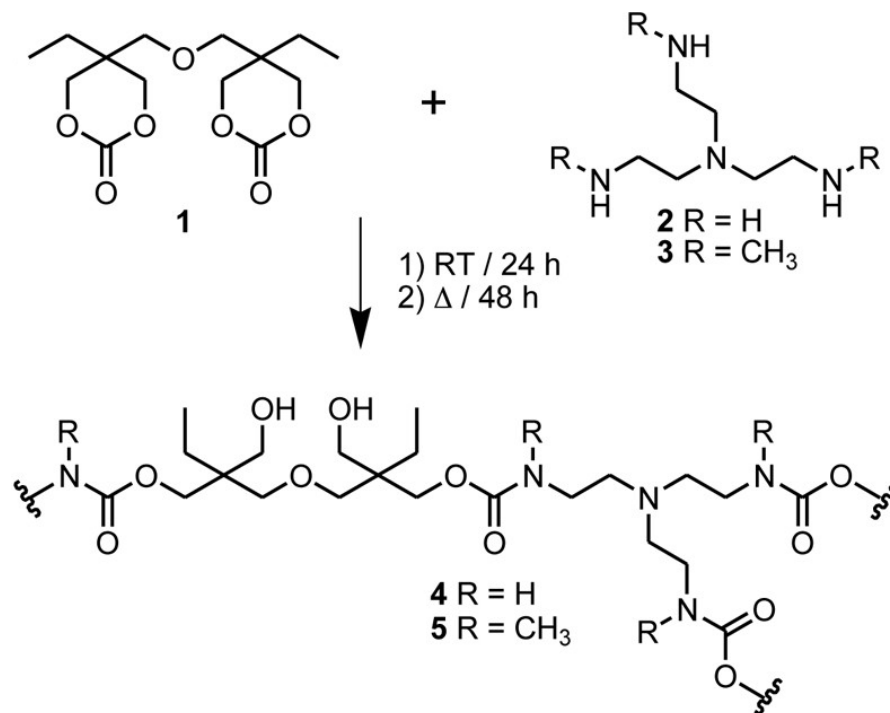
NaH

K<sub>2</sub>CO<sub>3</sub>

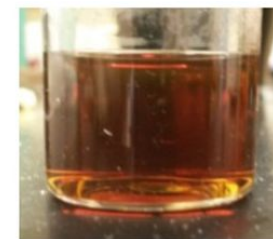
# Covalent Adaptable Networks

## Urethane CANs (II)

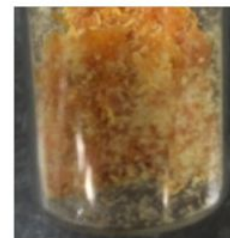
Example: *Isocyanate-Free Polyhydroxyurethanes*



Ph<sub>2</sub>O / 180 °C / 24 h



TEG / 180 °C / 4 h



160 °C / 8 h  
4 MPa



*J. Am. Chem. Soc.* **2015**, 137, 14019



# Covalent Adaptable Networks

## Further Linkages

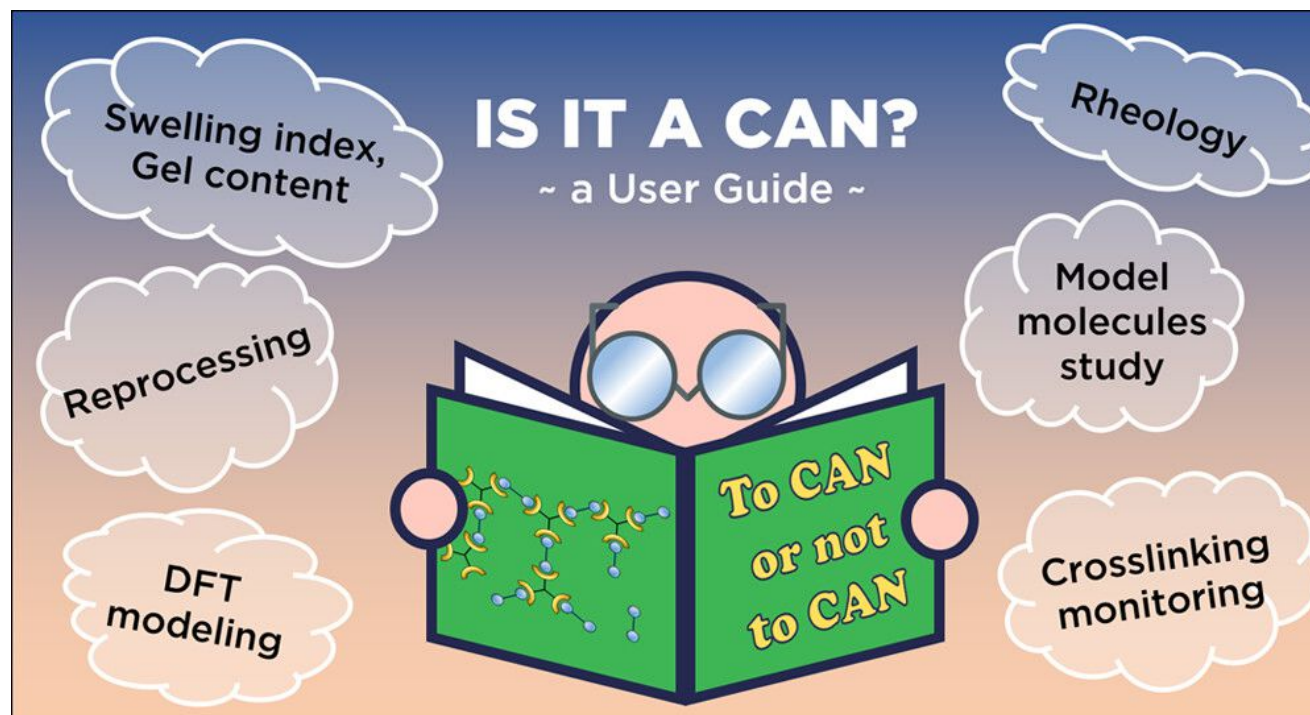
|                                                                             |      |
|-----------------------------------------------------------------------------|------|
| 4.8. Boronic Esters and Other Boron-Based Linkages                          | 1871 |
| 4.9. Acetals                                                                | 1874 |
| 4.10. Alkoxyamines                                                          | 1875 |
| 4.11. Hemiaminal and Triazine-Based Networks                                | 1876 |
| 4.12. Diketoenamines                                                        | 1877 |
| 4.13. Olefin/Olefin and Olefin/Imine Metathesis                             | 1878 |
| 4.14. New and Emerging CAN Linkages                                         | 1879 |
| 4.14.1. Aza–Michael Addition                                                | 1879 |
| 4.14.2. Trans- <i>N</i> -alkylation Using 1,2,3-Triazoliums and Other Salts | 1880 |
| 4.14.3. Diarylbibenzofuranone Cleavage and Recombination                    | 1880 |
| 4.14.4. Pyrazole-Urea Bond Exchange                                         | 1880 |
| 4.14.5. Nitroaldol Reactions                                                | 1881 |
| 4.14.6. Quaternary Anilinium Salts                                          | 1881 |
| 4.14.7. Reversible Nucleophilic Aromatic Substitution Exchange              | 1881 |

*Chem Rev.* **2022**, *13*, 370

# Covalent Adaptable Networks

## Characterization (I)

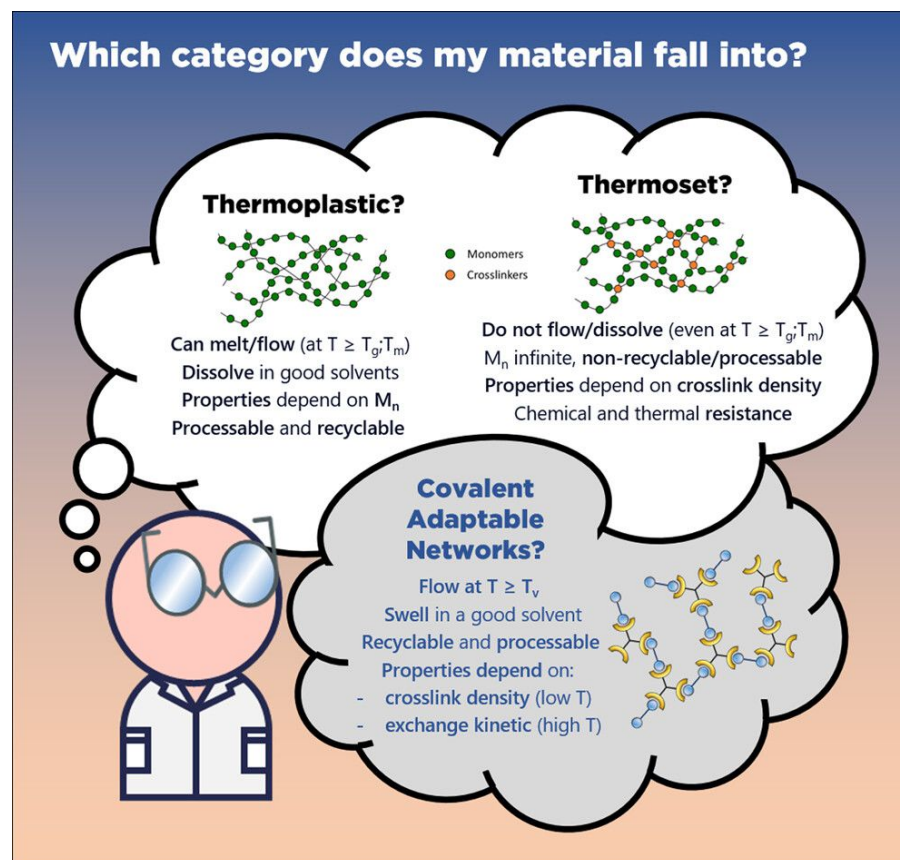
In principle, methods applicable to thermosets should be applied for CANs.



# Covalent Adaptable Networks

## Characterization (II)

The temperature-dependant measurements become even more relevant.



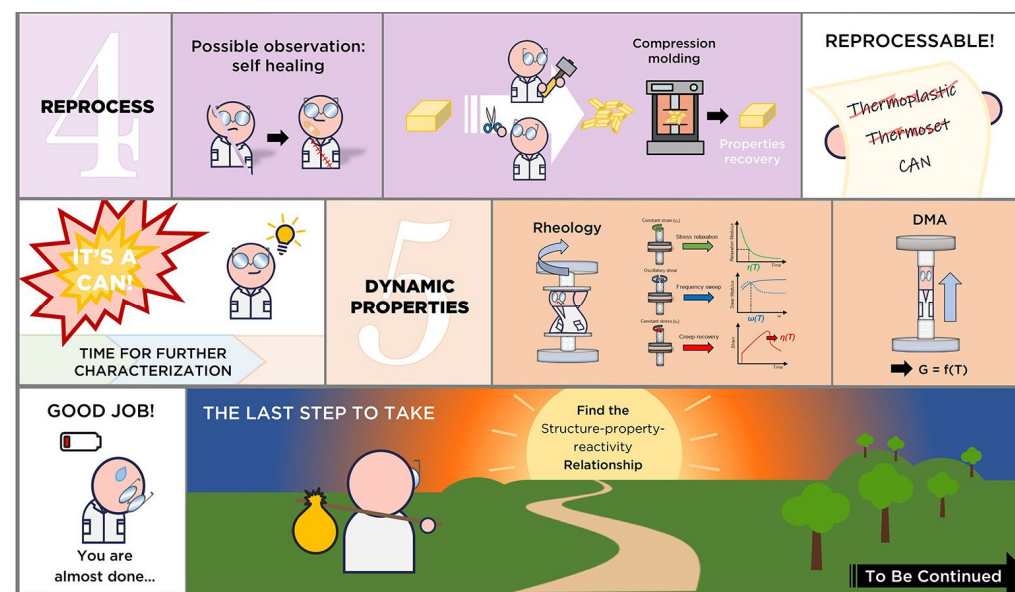
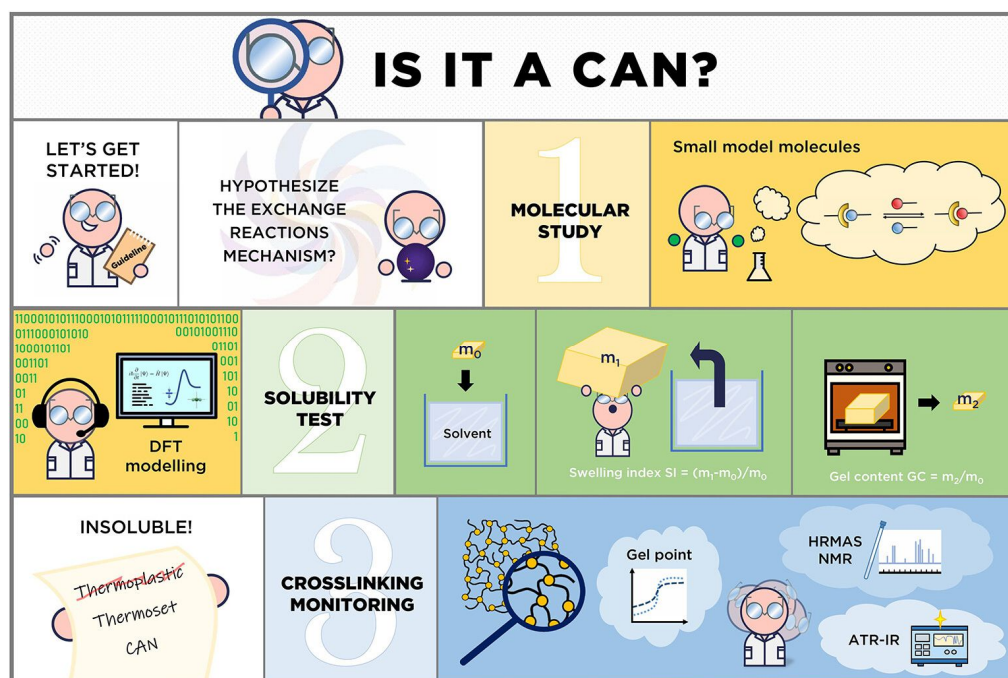
ACS Polym. Au 2025, 5, 214



# Covalent Adaptable Networks

## Characterization (III)

The real difference lies in the reprocessability.



# ***Crosslinked Polymers***

## Hydrogels

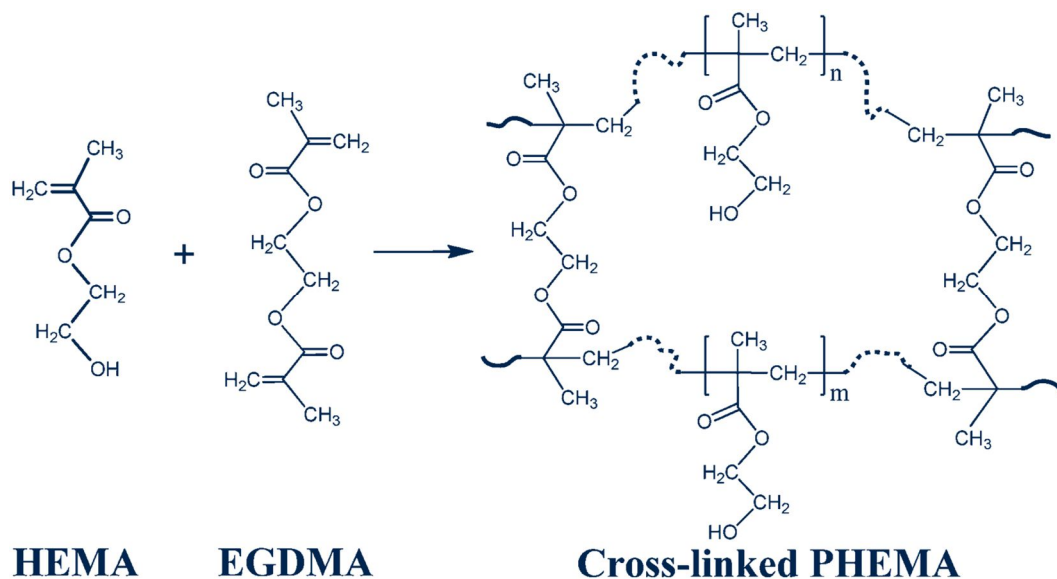
# Hydrogels

## Introduction

### Historical Perspective

Otto Wichterle and Drahoslav Lím developed poly(2-hydroxyethyl methacrylate) (PHEMA) hydrogels in the late 1950s–early 1960s, for contact lenses.

[www.tgw.cz](http://www.tgw.cz)



From Wichterle Family and  
the Contact Lens Museum

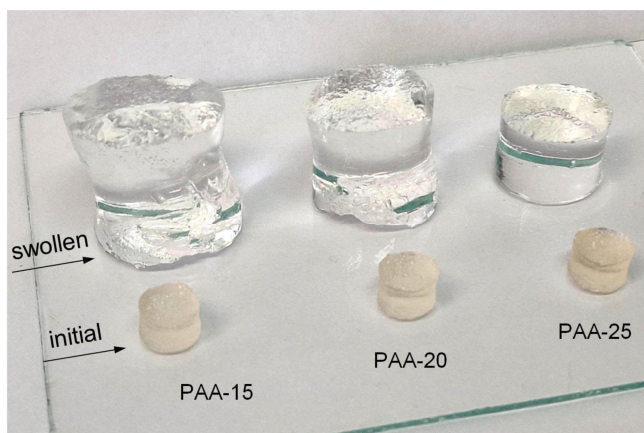


*Soft Matter* **2016**, 12, 1145  
*Substantia* **2022**, 6, 79

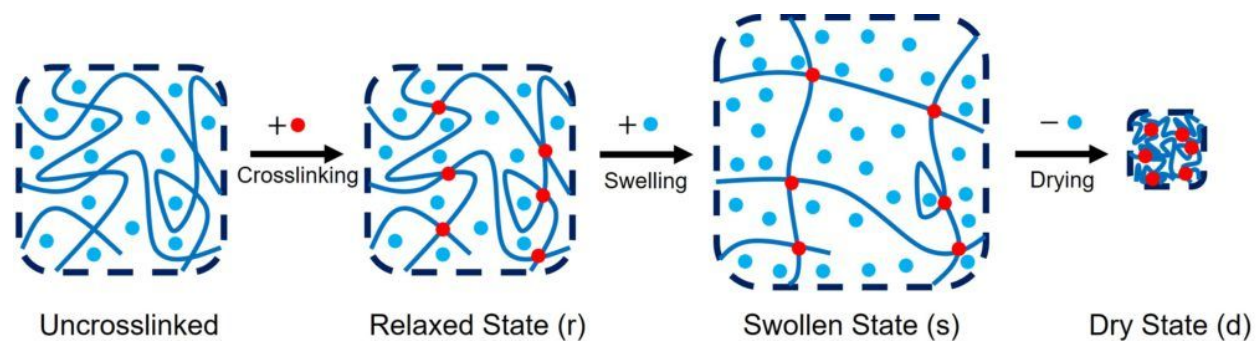
# Hydrogels

## Introduction

### From Crosslinked Polymers to Hydrogels



*J. Compos. Sci.* **2023**, *7*, 261



<https://hydrogeldesign.org/swelling/>

# Hydrogels

## Introduction

### Hydrogels among Soft Materials

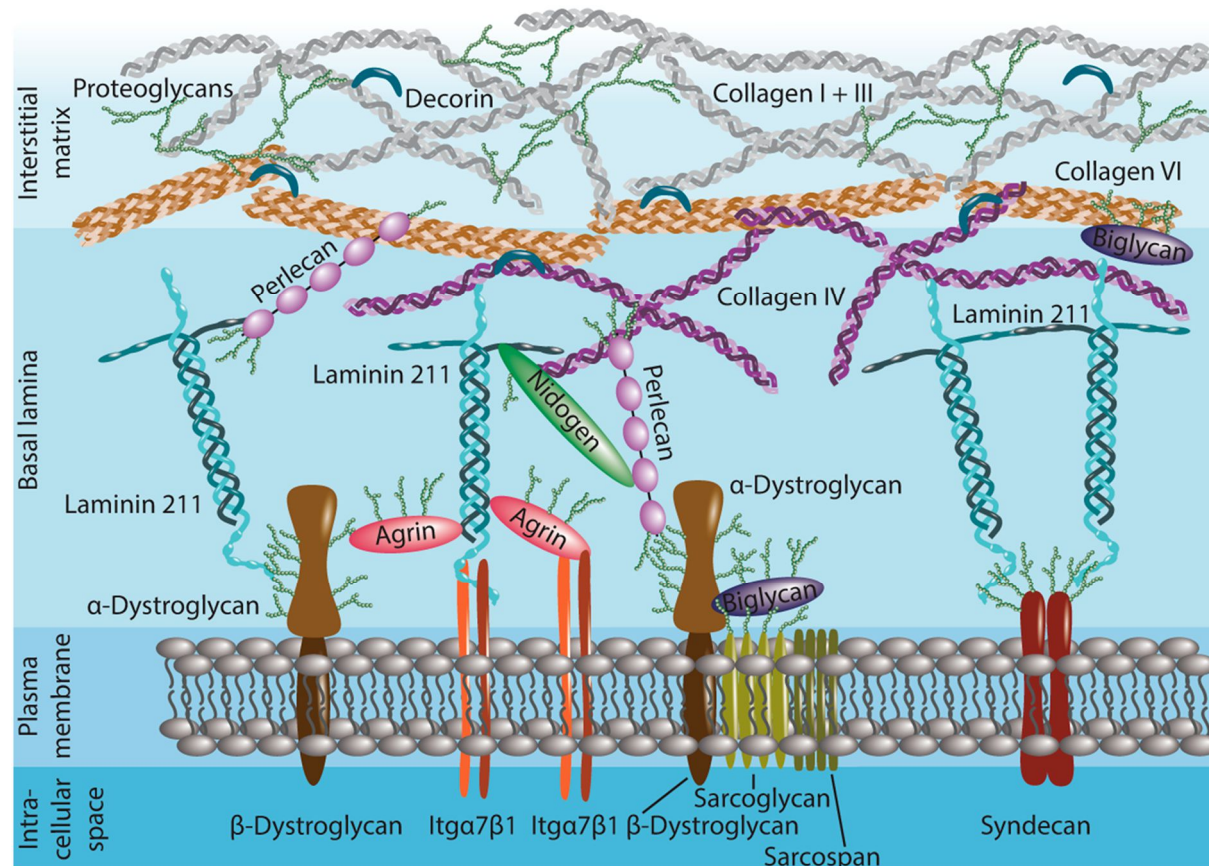
| Material         | Crosslinked | Continuous phase | Mechanical behavior |
|------------------|-------------|------------------|---------------------|
| Polymer solution | No          | Solvent          | Liquid              |
| Polymer melt     | No          | Polymer          | Viscous liquid      |
| Elastomer        | Yes         | Polymer          | Elastic solid       |
| Hydrogel         | Yes         | Water            | Soft solid          |
| Organogel        | Yes         | Organic solvent  | Soft solid          |



# Hydrogels

## Natural Hydrogels

### General Aspects



Front. Cell Dev. Biol. 2022, 10:1056523

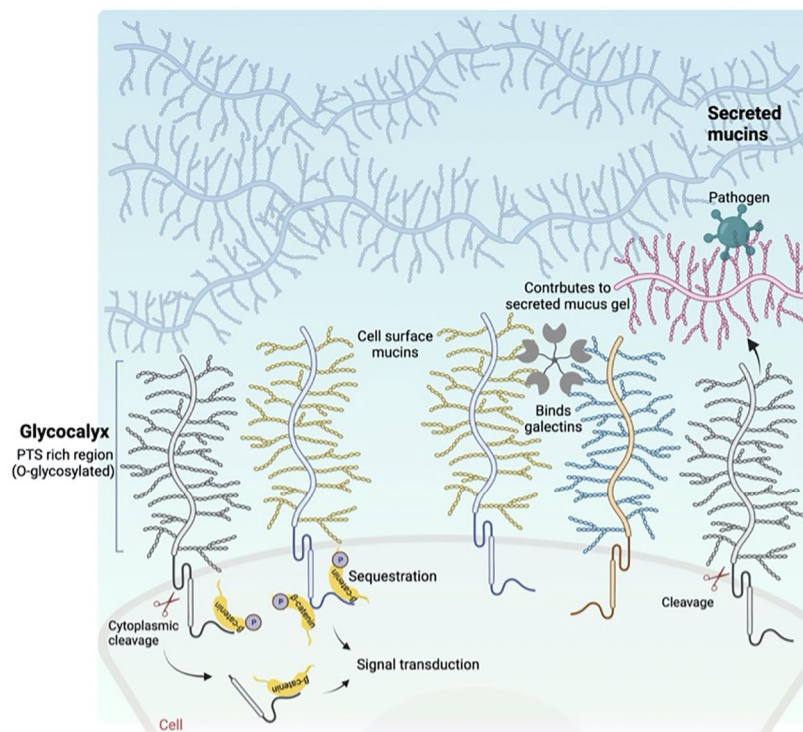
# Hydrogels

## Natural Hydrogels

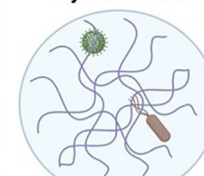
### Why Nature Uses Hydrogels

The key advantage of hydrogel-like structures in biology is the combination of:

- high diffusivity (nutrients, ions, metabolites)
- mechanical integrity (tissue stability)
- tunable viscoelasticity
- chemical functionality

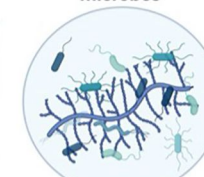


#### Physical Barrier



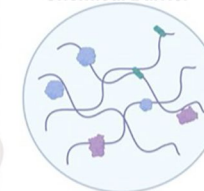
Forms a multimeric cross-linked network that acts as a gel-filtration system. Can bind pathogens and expel from the mucosa

#### Interaction with microbes



Able to segregate microbiota families, glycans are a nutrient resource for the commensals

#### Chemical Barrier



Sequesters antimicrobials and other proteins within the network. Providing additional protection and forming a chemical barrier

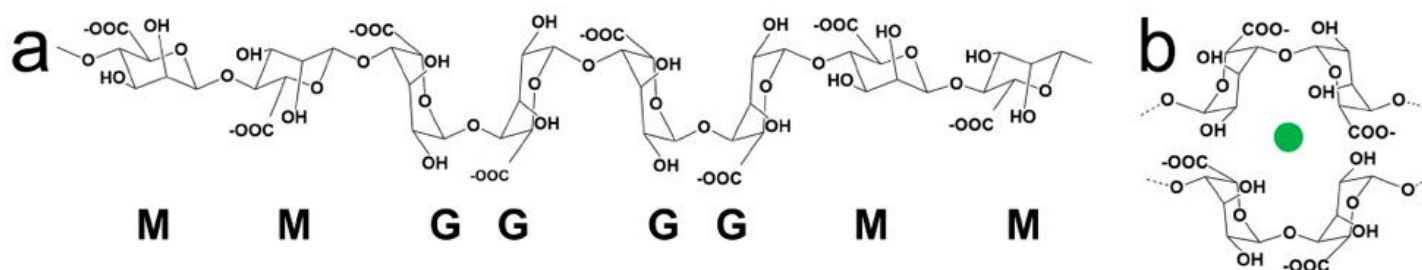
# Hydrogels

## Natural Hydrogels

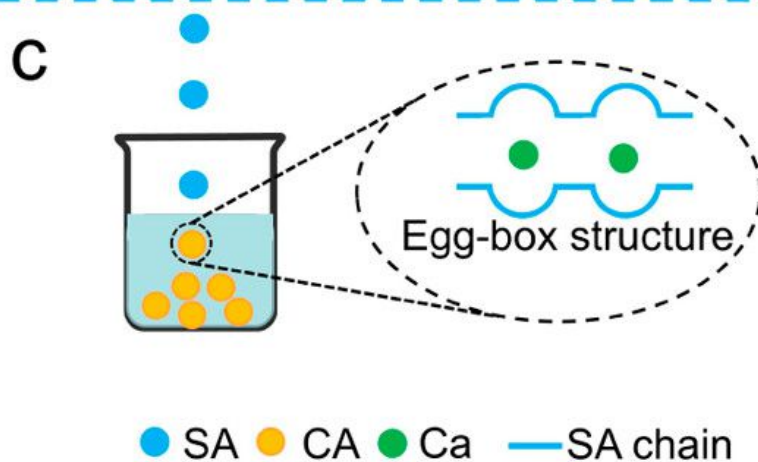
### Alginate: Ionically Crosslinked Polysaccharide

**M** mannuronic acid

**G** guluronic acid



● crosslinkers: divalent, trivalent and quadrivalent ions



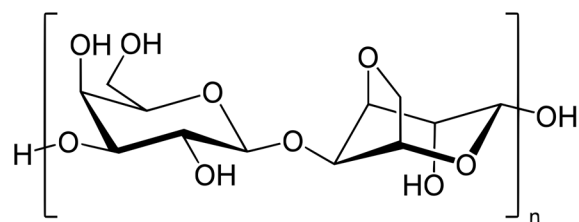
● SA ● CA ● Ca — SA chain

*Ind. Eng. Chem. Res.* **2023**, *62*, 11279

# Hydrogels

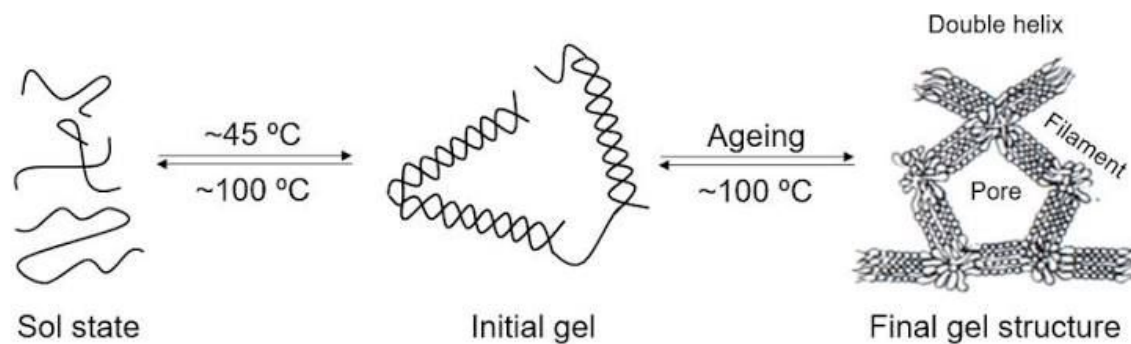
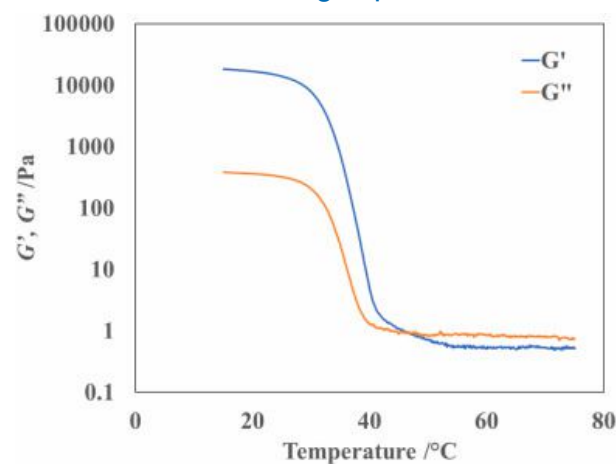
## Natural Hydrogels

### Agarose : Physical Gelation



D-galactose 3,6-anhydro-L-galactopyranose

cooling experiment



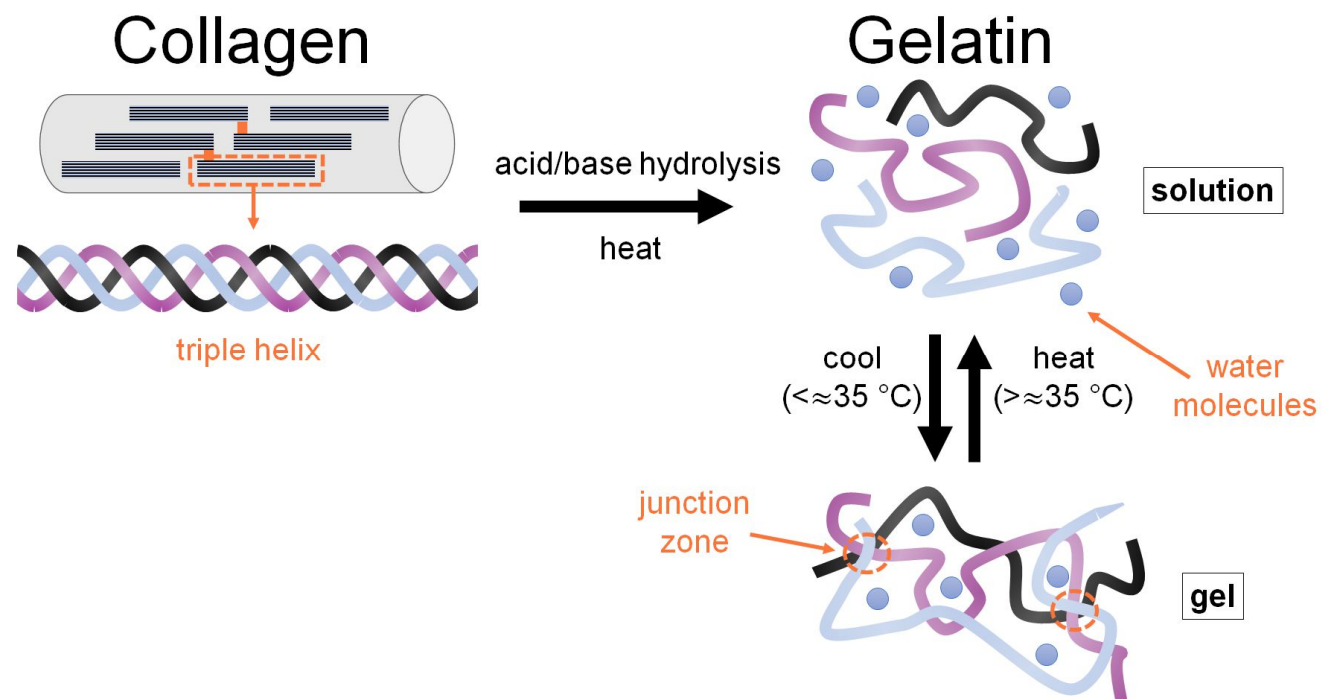
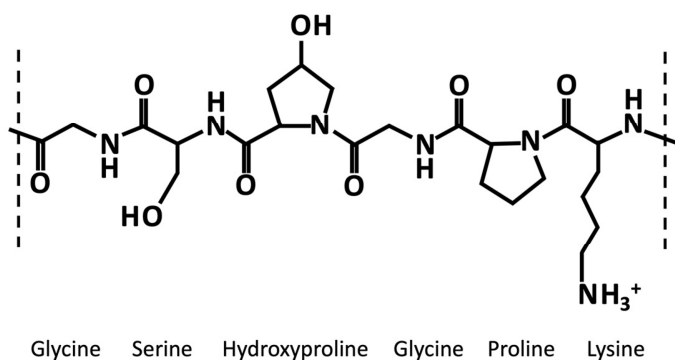
*Food Hydrocoll.* **2023**, 141, 108740  
*Express Polym. Lett.* **2017**, 11, 244



# Hydrogels

## Natural Hydrogels

### Gelatin: Denatured Collagen Network



*Pharmaceutics* **2023**, *15*, 1499

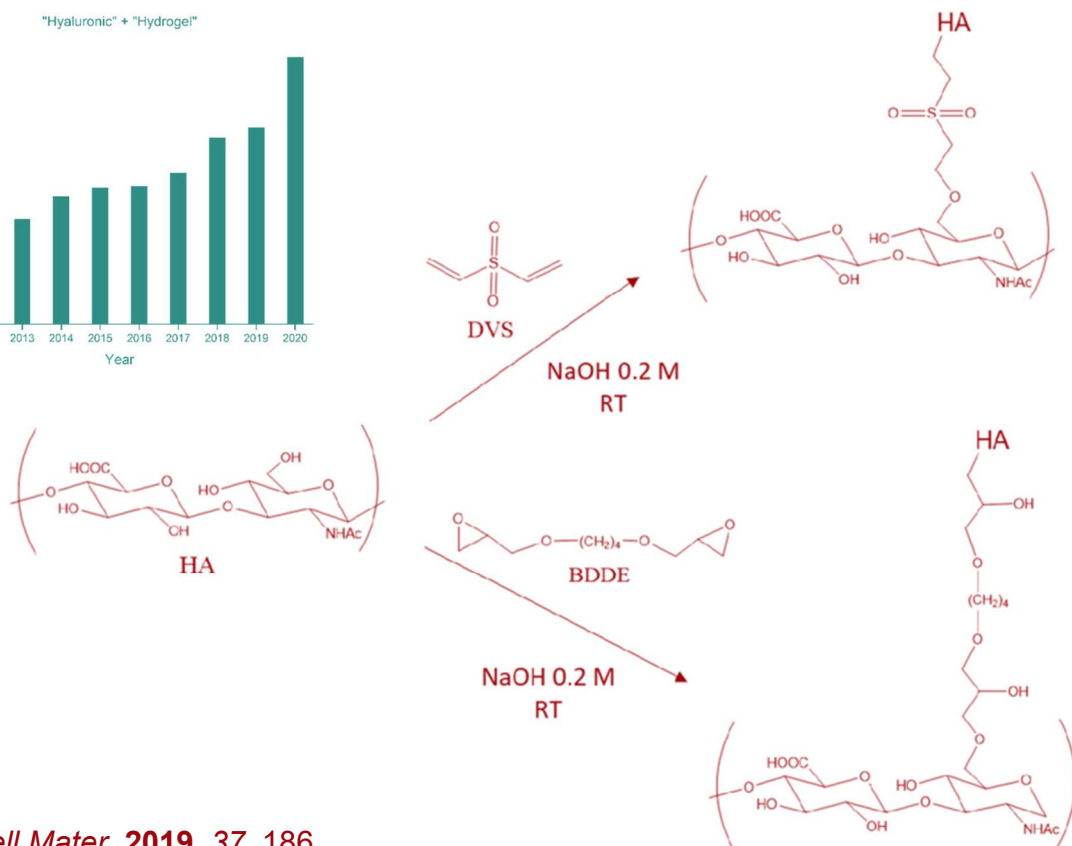
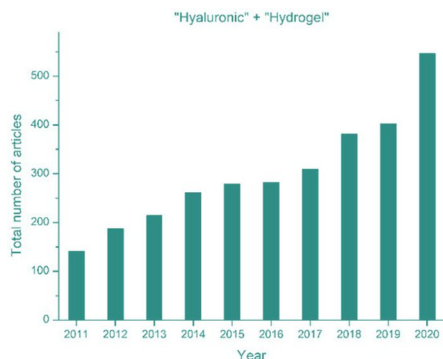
<https://chembam.com/resources-for-students/the-chemistry-of/gelatin/>



# Hydrogels

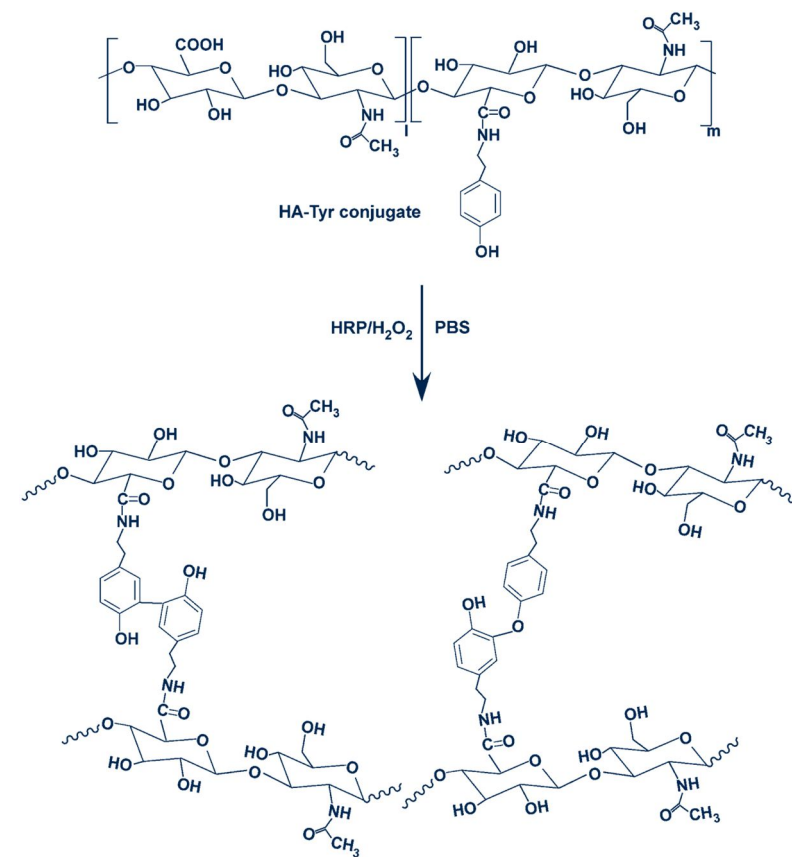
## Natural Hydrogels

### Hyaluronic Acid: Polyelectrolyte Hydrogel



*Eur. Cell Mater.* **2019**, 37, 186

*Biomedicines* **2021**, 9, 1113



*Soft Matter* **2008**, 4, 880